Trade-off between Privacy and Explainability in House Allocation problems

Sarah Lamik[1] , **Alexandre Faure**[1] , **Milan Garnier**[1] , **Anaëlle Wilczynski**[2] , **Wassila Ouerdane**[2]
[1]Paris-Saclay University, CentraleSupélec
[2]Paris-Saclay University, CentraleSupélec, Laboratory of Mathematics and Computer Science (MICS),
{alexandre.faure, milan.garnier, sarah.lamik}@student-cs.fr, {anaelle.wilczynski, wassila.ouerdane}@centralesupelec.fr

Abstract

In the study of multi-agent resource allocation with indivisible goods, ensuring algorithmic transparency is critical for user trust and system acceptance. When agents are assigned items in settings like in the house allocation problem, they frequently ask counterfactual questions (e.g., "Why was I not allocated a better item?") to verify the fairness of the outcome. Providing such explanations, however, inherently risks disclosing the private preference profiles of other participating agents, introducing a fundamental conflict between explainability and preference privacy.

By encoding the allocation, preferences, and fairness axioms into Boolean Satisfiability (SAT) formulas, we answer local counterfactual queries by extracting Minimal Unsatisfiable Subsets (MUSes). Furthermore, we formalize quantitative metrics to measure the exact degree of data disclosure induced by an explanation, defined by the reduction of the querying agent's posterior compatible set of preference profiles. Experiments are conducted on small instances ($n \leq 4$) due to the exponential cost of exact profile enumeration. Alongside these metrics we propose a framework for navigating the trade-off between generating concise explanations and preserving the privacy of agents in fair division problems.

Keywords: XAI, Preference Privacy, Fair Division, SAT, Counterfactual Explanations

1 Introduction

House allocation problems arise in many practical settings where indivisible items must be assigned to agents, such as project or task allocation in workplaces and schools, course selection in universities, or seating arrangements at events. In these contexts, the final allocation is typically observable by all agents, raising natural questions about its fairness and the reasons behind it.

Providing explanations for such allocations is therefore essential to ensure transparency and acceptance. However, explaining an allocation often requires access to agents' preferences, which may contain sensitive information. Revealing too much of this preference profile can expose disagreements or private judgments, creating a tension between *explainability* and *privacy*.

In this paper, we study how to generate explanations for house allocation outcomes while limiting the amount of preference information disclosed. We assume that the given allocations satisfy *rank(-k)-envy-freeness*, a relaxation of envy-freeness that can always be achieved and computed efficiently. Building on this assumption, we develop a framework for producing concise and informative explanations while assessing the level of data disclosure.

Our approach aims to bridge the gap between fairness justification and preference privacy, providing tools to reason about the trade-off between explanation quality and the amount of information revealed.

2 Related Work

2.1 Fair Allocation and Equity Notions

Fair allocation is a central topic in computational social choice, with standard references such as Brandt *et al.* [2016]. Envy-freeness is a canonical fairness notion, but in house allocation problems, where each agent receives a single indivisible item, envy-free allocations may fail to exist. This has led to the study of relaxations, notably local envy-freeness and rank-based notions Beynier *et al.* [2019]; Belahcène *et al.* [2021]; Ramezanian and Feizi [2020]; Kojima and Ünver [2014].

In particular, rank-envy-freeness and its generalisations provide tractable fairness guarantees that are always achievable and computable in polynomial time Beynier *et al.* [2024]. These properties make them especially suitable for practical systems, including those requiring post-hoc justification.

2.2 Explainability in Fair Allocation

Explainability has become a key requirement in AI, especially for decision-making systems Arrieta *et al.* [2020]; Gunning *et al.* [2019]; Mohseni *et al.* [2021]. In this context, we focus on post-hoc, instance-specific explanations of fixed outcomes.

In computational social choice, justification has been formalised as deriving an outcome from a minimal set of axiom instances Boixel and Endriss [2020]. This approach has been made computationally viable through graph-based and SAT-backed methods Nardi *et al.* [2022]. These techniques enable the extraction of concise, logically grounded explanations.

Recent work applies this framework to fair allocation and matching. In particular, Beynier *et al.* [2024] explain the non-existence of locally envy-free allocations, while Knapp [2022]; Ouerdane *et al.* [2025] study explanation in matching and roommate settings. However, these works primarily focus on infeasibility explanations or settings with different preference structures, leaving the explanation of given allocations comparatively underexplored.

2.3 MUS-Based Explanations and SAT Approaches

SAT-based encodings are widely used to reason about fairness constraints Brandt *et al.* [2016]. Minimal Unsatisfiable Subsets (MUSes) provide minimal certificates of inconsistency and form the basis of explanation generation Liffiton and Sakallah [2008]; Liffiton *et al.* [2016].

These techniques have recently been adapted to social choice settings Nardi *et al.* [2022]; Beynier *et al.* [2024], enabling structured and minimal explanations of fairness-related properties. However, a key limitation of existing MUS-based approaches is their reliance on uncontrolled preference disclosure.

Gap. While preference privacy and elicitation have been studied independently, their interaction with explainability in fair allocation remains largely unexplored. Existing approaches assume complete access to agents' preferences, which is often unrealistic or undesirable in practice. In particular, there is no framework for controlling the amount of preference information revealed in explanations, nor for evaluating the trade-off between explanation quality and data disclosure.

Our work addresses this gap by developing an explanation framework for house allocation that explicitly accounts for limited preference disclosure, enabling privacy-aware justifications of allocation outcomes.

3 Background and Preliminaries

3.1 House allocation problem (formal definition)

For any positive integer k, we denote $[k] = \{1, ..., k\}$. We are considering a set of agents $\mathcal{A} = [n]$, with n a positive integer, and $\mathcal{O} = \{o_1, ..., o_n\}$ a set of objects (or items). Each agent $i \in \mathcal{A}$ expresses their preferences as strict ordinal preferences over all items as a linear order $\succ_i$ over $\mathcal{O}$. The set of all agents' preference orders is called a preference profile and is denoted by $\succ$. A house allocation instance is given by the triplet $H = (\mathcal{A}, \mathcal{O}, \succ)$. The set of all possible house allocation problem is denoted $\mathcal{H}$. An allocation on an instance H is denoted $\sigma : \mathcal{A} \to \mathcal{O}$ where $\sigma(i)$ is the object assigned to agent i.

Based on an instance $H = (\mathcal{A}, \mathcal{O}, \succ)$, for each agent $i \in \mathcal{A}$ and each object $o \in \mathcal{O}$, we denote $r_i(o) \in [n]$ the rank of object o in agent i's preference order. Formally, $r_i(o) = |\{o' \in \mathcal{O}/o' \succeq_i o\}|$. In addition, we will denote $top(o)$ for every object o as the set of agents who ranked o first, *i.e.* $top(o) = \{i \in \mathcal{A}/r_i(o) = 1\}$. Finally, T will be the subset of objects ranked first at least once: $T = \{o \in \mathcal{O}/|top(o)| \geq 1\}$.

3.2 Equity notions (r-EF , r_k-EF)

In this article, we will focus on *fair allocations*. To assess fairness, we will consider the *rank-envy-freeness*. An allocation σ is said to be *rank-envy-free* if no agent envy another one by taking into account the ranks of the objects for both agents.

Definition 3.1 (Rank-envy-freeness (r-EF)). *An allocation σ is said to be rank-envy-free (r-EF) if for all agents $i, j \in \mathcal{A}$, $\sigma(i) \succ_i \sigma(j)$ or $r_j(\sigma(j)) \leq r_i(\sigma(j))$.*

Proposition 3.2. *For any instance $H = (\mathcal{A}, \mathcal{O}, \succ)$, there exists an allocation σ that is rank-envy-free.*

Algorithm 1: Polynomial-time algorithm for Serial Dictatorship like r-EF allocation

```
Let A ← 𝒜
Let O ← 𝒪
Let i ← 1
while A ≠ ∅ and O ≠ ∅ do
    Let
    r_min ← min{r ∈ [n]/∃(i,o) ∈ A×O, r_i(o) = r};
    Let candidates ← {(i,o) ∈ A × O/r_i(o) = r_min};
    Let (i,o) ← pick_random_candidate(candidates);
    Let σ_i ← (i,o);
    Set A ← A \ {i};
    Set O ← O \ {o};
    Set i ← i + 1;
return {σ_1, ..., σ_n};
```

Definition 3.3 (Rank-k-envy(r_k-EF)). *Agent i **rank-k-envies** agent j under allocation σ and preference profile $\succ$ if and only if: $\sigma(j) \succ_i \sigma(i)$ and $r_j(\sigma(j)) > \min\{r_i(\sigma(j)), k\}$*

Definition 3.4 ((Rank-k-envy-freeness (r_k-EF)). *An allocation σ is **rank-k-envy-free** with respect to profile $\succ$ if no agent rank-k-envies any other:*

$\forall i, j \in \mathcal{A}$, $\sigma(i) \succ_i \sigma(j)$ *or* $r_j(\sigma(j)) \leq \min\{r_i(\sigma(j)), k$

Intuitively, σ is r_k-EF if whenever i envies j, it must be the case that j both values her object more than i does and ranks it within her top-k. Note that :

- The r-EF and the r_{n-1}-EF are equivalent for an instance of size n.
- r_k-EF $\implies$ $r_{k'}$-EF for every $k < k'$ and r_{n-1}-EF $\Leftrightarrow$ r-EF

3.3 SAT encoding of house allocation and equity

Matching consistency

The following axioms ensure that the matching σ is a bijection from $\mathcal{A}$ to $\mathcal{O}$.

- Each agent receives *at least* one object :

$$\forall i \in \mathcal{A} : \phi_H^{\geq 1,\mathcal{A}}(i) := \bigvee_{j \in \mathcal{O}} x_{i,o}$$

- Each agent receives *at most* one object :

$$\forall i \in \mathcal{A}, o, o' \in \mathcal{O}^2, o \neq o' : \phi_H^{\leq 1,\mathcal{A}}(i,o,o') := \neg x_{i,o} \vee \neg x_{i,o'}$$

- Each object gets matched to *at least* one agent :

$$\forall j \in \mathcal{O} : \phi_H^{\geq 1,\mathcal{O}}(j) := \bigvee_{i \in \mathcal{A}} x_{i,o}$$

- Each object gets matched to *at most* one agent :

$$\forall o \in \mathcal{O}, i, j \in \mathcal{A}^2, i \neq j : \phi_H^{\leq 1,\mathcal{O}}(o,i,j) := \neg x_{i,o} \vee \neg x_{j,o}$$

Those four types of clauses are not all necessary to encode the matching consistency. Indeed, $(\phi_H^{\geq 1,\mathcal{A}}(i))_{i \in \mathcal{A}}$ and $(\phi_H^{\leq 1,\mathcal{O}}(o,i,j))_{o,i,j \in \mathcal{O} \times \mathcal{A}^2}$ would be sufficient. The other axioms allow redundancy which can help make explanations shorter as shown by Beynier *et al.* [2024].

r-EF SAT formulation

We can express the r-EF property under several forms, including the direct, restriction or characterization formulation Ouerdane *et al.* [2025]. As the characterizations has been found to be the most efficient in the fair Roommate Matching setup to obtain short explanations, we are going to consider the following SAT formulation of this property for an instance H:

$$\phi_H^{ref} := \bigwedge_{o \in T} \phi_H^{top}(o) \wedge \bigwedge_{o \in \mathcal{O} \setminus T} \bigwedge_{\ell=2}^{n-1} \bigwedge_{\substack{i \in \mathcal{A} \\ r_i(o) = \ell}} \phi_H^{rk}(o,\ell,i)$$

$$\text{with} \quad \phi_H^{top}(o) := \bigvee_{i \in top(o)} x_{i,o}$$

$$\text{and} \quad \phi_H^{rk}(o,\ell,i) := \bigvee_{\substack{o' \in \mathcal{O} \setminus \{o\} \\ r_i(o') < \ell}} x_{i,o'} \vee \bigvee_{\substack{j \in \mathcal{A} \setminus \{i\} \\ r_j(o) \leq \ell}} x_{j,o} \vee x_{i,o}$$

3.4 Explanation framework: questions and formal definition

We adopt a contrastive explanation framework rooted in the work and taxonomy developed in Miller [2019] and in Lipton [1990]. In this paradigm, an agent i^* does not ask an explanation for an allocation *in itself*, but rather relative to some preferred alternative: the question takes the contrastive form *"Why did I receive $p^* = \sigma(i)$ rather than q^*?"*. Following Lipton [1990], we call the actual outcome the **fact** and the preferred alternative the **foil**. The explanation must then identify a structural difference between the world in which i^* receives $\sigma(i)$ and the counterfactual world in which i receives q^*, making the latter **impossible under the axioms governing the mechanism**. This framing is well-motivated empirically: Miller [2019] establishes, through a systematic survey of the social science literature, that human explanatory demands are overwhelmingly contrastive in nature. This observation is particularly salient in fair allocation: a dissatisfied agent does not simply wish to understand the allocation mechanism in general, but to understand why *her specific* counterfactual request cannot be granted.

Notation. We denote by $i^* \in \mathcal{A}$ the *querying agent* and by $q^* \in \mathcal{O}$ the *object of the query*, satisfying $q^* \succ_{i^*} \sigma(i^*)$. The formal object underlying an explanation is a *Minimal Unsatisfiable Subset* (MUS) Liffiton and Sakallah [2008]: given an unsatisfiable CNF formula Ψ, a MUS is a subset $\psi \subseteq \Psi$ such that ψ is unsatisfiable yet every proper subset of ψ is satisfiable. MUSes have been used as the formal backbone of explanations in collective decision problems, including house allocation Beynier *et al.* [2024] and roommate matching Ouerdane *et al.* [2025].

Definition 3.5 (Explanation Beynier *et al.* [2024])**.** *Let $\mathcal{E}$ denote the set of all possible explanations. An explanation $Expl \in \mathcal{E}$ for the query (i^*, q^*) is a set of axioms $Expl = \{A_1, \ldots, A_m\}$ whose associated clauses form a MUS of the augmented formula $\Psi(H,Q)$, and which contains the counterfactual clause encoding the query.*

By minimality of the MUS, every axiom in *Expl* is *necessary*: removing any single axiom restores satisfiability. The explanation is therefore both *sound* since it correctly witnesses the impossibility of the counterfactual and *non redundant*. As shown in Ignatiev et al. [2020], this abductive structure is formally dual to contrastive explanation: the foil q^* cannot be achieved precisely because the conjunction of axioms in *Expl* forbids it.

For a given query Q and instance H, the formula $\Psi(H,Q)$ may admit *multiple* MUSes, each yielding a distinct explanation. We denote by $\mathcal{M}(H,Q)$ the set of all such MUSes. As established in Beynier *et al.* [2024], the choice of MUS has a material impact on the clarity and conciseness of the resulting explanation, motivating the introduction of metrics over $\mathcal{M}(H,Q)$, which we develop in the following subsection.

3.5 MUS: definition and role in explanations

From MUS to explanation

MUSes are particularly useful to extract a minimal subset of clauses that illustrate that a SAT problem is unsatisfiable. Such technique has already been used to explain why locally-envy-free allocations don't exist for a given instance in house allocation problem Beynier *et al.* [2024] or in roommate matching Ouerdane *et al.* [2025].

We briefly recall the explanation framework introduced by Beynier *et al.* [2024], which derives structured explanations from a Minimal Unsatisfiable Subset (MUS). Let ψ be an unsatisfiable set of clauses. The key object is the *satisfiability implication graph* (SIG), a directed bipartite graph $G_\psi = (V^\psi, E^\psi)$ encoding logical dependencies between clauses and variables.

The node set V^ψ is partitioned into *variable nodes*, corresponding to the variables appearing in ψ, and *clause nodes*, corresponding to the clauses of ψ. Two special nodes, $\top$ and $\perp$, represent tautological truth and contradiction, respectively. Each clause is rewritten as an implication from a conjunction of positive literals (or $\top$) to a disjunction of positive literals (or $\perp$). Accordingly, for each clause $C \in \psi$, edges are added from variable nodes in the implicant part of C to the clause node of C, and from this clause node to the variable nodes in its implied part.

To reason over this structure, an *activation function* $v : V^\psi \to \{0,1\}$ is defined, where $v(x) = 1$ indicates that node x is active. Intuitively, active variable nodes correspond to literals set to true, while active clause nodes represent clauses whose antecedent is satisfied and whose consequent must therefore hold. Starting from an initial activation state where only $\top$ is active, activation propagates through the graph according to the implication structure: a clause becomes active when all its predecessors are active, and its activation forces at least one of its successors to become active.

An *activation path* is a sequence of such states obtained by successive propagation steps. Beynier *et al.* [2024] show that, if ψ is unsatisfiable, every maximal activation path eventually activates $\perp$. This process can be explored using a depth-first search over activation states.

Textual explanations are extracted by tracing an activation path from $\top$ to $\perp$ and verbalising the semantics of nodes as they become active. Each node is associated with a domain-specific interpretation (e.g., preference or allocation constraints), so that the sequence of activations forms a step-by-step explanation of the contradiction. Since ψ is a MUS, the resulting explanation is both *sound* (it derives the contradiction) and *minimal* (it contains no redundant constraints), yielding a concise and logically grounded justification.

The table 1 gives the textual translation of each node.

Node	**Semantics**
x_{io}	Suppose agents i receives object o.
$\phi_H^{\geq 1,\mathcal{A}}(i)$	Agent i must receive at least one object.
$\phi_H^{\geq 1,\mathcal{O}}(o)$	Object o must be matched with at least one agent.
$\phi_H^{\leq 1,\mathcal{A}}(i,o,o')$	Agent i cannot receive both o and o'.
$\phi_H^{\leq 1,\mathcal{O}}(o,i,j)$	Object o cannot be assigned to both i and j.
$\phi_H^{top}(o)$	Since object o is ranked first by some agents, o must be matched with an agent who ranks it first.
$\phi_H^{rk}(o,\ell,i)$	Object o is ranked by agent i at rank ℓ; to avoid rank-envy, if i does not receive an object it prefers to o, then o must be matched with an agent who ranks it at a rank smaller or equal to ℓ.

Table 1: Semantics of nodes in our satisfiability implication graphs

Metrics on MUSes space

As shown in the paper from Beynier *et al.* [2024], the MUS chosen to extract an explanation have a considerable impact on the clarity and brevity of the explanation.

We will thus introduce *metrics* to assess some quality of an explanation (*wrt.* conciseness or privacy). A metric m applied to (H,Q) is a function $m_{H,Q} : \mathcal{M}(H,Q) \to \mathbb{R}^+$ to minimize.

Explanation clarity and conciseness metrics

To assess the clarity and conciseness of an explanation, we introduce a set of metrics that quantify both structural and computational aspects of the explanation. These metrics are designed to capture the complexity and comprehensibility of the explanation, inspired by the framework of satisfiability implication graphs.

Before delving into graph-based metrics, we consider a set of simple yet informative metrics that directly relate to the components involved in the explanation:

- **Number of agents (#agents):** The count of agents whose preferences or constraints are explicitly referenced in the explanation.
- **Number of objects (#objects):** The count of objects mentioned in the explanation.
- **Number of clauses (#clauses):** The total number of logical clauses (*e.g.* $\phi_H^{\geq 1,\mathcal{A}}(1)$) explicitly mentioned in the explanation.
- **Number of variables (#variables):** The total number of distinct variables (*e.g.* x_{1,o_1}) that appear in the explanation.

These metrics provide a high-level overview of the explanation's scope and complexity, serving as a baseline for evaluating its clarity.

To capture the structural intricacies of the explanation, we can also take advantage of the satisfiability implication graph developed earlier and use the metrics introduced by Beynier *et al.* [2024]. Let ψ be a minimally unsatisfiable subset (MUS) of the instance, and let H_ψ denote its associated satisfiability implication graph. An explanation is constructed by performing a depth-first search over all possible activation states of H_ψ.

Definition 3.6 (Explanation Length (L_ψ))**.** *For a given MUS ψ, the **explanation length** L_ψ is defined as the total number of activation states explored during the search, i.e.,*

$$L_\psi = |S_\psi|,$$

where S_ψ is the set of all activation states associated with ψ.

Definition 3.7 (Explanation Depth (d_ψ))**.** *The **explanation depth** d_ψ quantifies the longest path in the activation graph required to reach a contradiction. Formally,*

$$d_\psi = \max_{v \in P_\psi} |v|,$$

where P_ψ is the set of all possible activation paths in H_ψ.

Definition 3.8 (Explanation Breadth (B_ψ))**.** *The **explanation breadth** B_ψ measures the number of distinct activation paths required to fully explore the MUS ψ, i.e.,*

$$B_\psi = |P_\psi|.$$

These graph-based metrics provide a nuanced understanding of the explanation's complexity, reflecting both its structural depth and the number of alternative paths that must be considered to reach a contradiction.

4 Theoretical Contributions

4.1 Extended explanation framework

We consider an agent i^* who receives object $p^* = \sigma(i^*)$ and is dissatisfied with this outcome. She may query the system with a **counterfactual question** about alternative allocations. In this section, the agent and the objects considered in the question are denoted with *.

$\mathrm{r_k}$-EF direct SAT formulation

The direct SAT formulation of the $\mathrm{r_k}$-EF is:

$$\phi_H^{\mathrm{r_k\text{-}EF\text{-}dir}} := \bigwedge_{\substack{i,j\in\mathcal{A}\\ i\neq j}} \bigwedge_{\substack{o,o'\in\mathcal{O}\\ o'\succ_i o\\ r_j(o')>\min(k,r_i(o'))}} \phi_H^{\mathrm{dir}}(i,j,o,o')$$

where $\forall i,j,o,o'$ as above:

$$\phi_H^{\mathrm{dir}}(i,j,o,o') := \neg x_{i,o} \vee \neg x_{j,o'}$$

Condition: Agent i prefers o' over o AND agent j ranked o' outside their top-k (or outside top-rank of o' from i's perspective).

$\mathrm{r_k}$-EF characterization for SAT formulation

Let's define the characterisation CNF for the $\mathrm{r_k}$-EF for $k > 0$:

$$\phi_H^{\mathrm{r_k\text{-}EF}} := \bigwedge_{o\in T} \phi_H^{top}(o) \wedge \bigwedge_{o\in\mathcal{O}\setminus T} \bigwedge_{\ell=2}^{n-1} \bigwedge_{\substack{i\in\mathcal{A}\\ r_i(o)=\ell}} \phi_H^{rk,k}(o,\ell,i)$$

with $\quad \phi_H^{rk,k}(o,\ell,i) := \bigvee_{\substack{o'\in\mathcal{O}\setminus\{o\}\\ r_i(o')<\ell}} x_{i,o'} \vee \bigvee_{\substack{j\in\mathcal{A}\setminus\{i\}\\ r_j(o)\leq\min(\ell,k)}} x_{j,o} \vee x_{i,o}$

Note that for $k = 0$, the only $\mathrm{r_0}$-EF allocation (if it exists) is the one that assigns to each agent their preferred object.

Furthermore, $\phi_H^{rk}(o,\ell,i)$ corresponds to the special case $\phi_H^{rk,k}(o,\ell,i)$ with $k = n-1$.

Table 2 provides the semantic of clauses $\phi_H^{rk,k}(o,\ell,i)$.

Types of questions

In this paper, we distinguish between two types of questions that might be asked as an extension of the work done by Belahcene *et al.* [2018]. We define *global questions* are questions related to the entire allocation for a given instance H and an allocation σ and aims to assess procedural regularity. Whereas *local questions* are related to an instance H with regards to an allocation σ and concerns only a single agent $i \in \mathcal{A}$.

In this paper, we will only consider local questions that we further divide into two types:

Definition 4.1 (Local Question Type 1 (Q^1_{loc})). *"Why did I not receive object q^* that I prefer?" where $r_{i^*}(q^*) < r_{i^*}(p^*)$. One corresponding CNF formulation is $Q^1_{loc} = x_{i^*,q^*}$.*

Definition 4.2 (Local Question Type 2 (Q^2_{loc})). *"Why did I not receive an object better than p^*?". One corresponding CNF formulation is $Q^2_{loc} = \bigvee_{\substack{q\in\mathcal{O}\\ q\succ_{i^*} p^*}} x_{i^*,q}$*

These queries are encoded in the SAT solver. If the resulting formula becomes **unsatisfiable**, a **MUS** is extracted.

We can naturally extend the semantics of these clauses as is:

Node	Semantics
$\phi_H^{rk,k}(o,\ell,i)$	To avoid rank-k-envy about object o, and because agent i ranked it at position ℓ, either i should have a better object, or o should be affected to another agent who ranked it lower than k and ℓ.
$Q^1_{loc}(i^*,q^*)$	Agent i^* asks if they can obtain object q^*.
$Q^2_{loc}(i^*,\ell)$	Agent i^* asks if they can obtain an object they ranked strictly better than rank ℓ.

Table 2: Extension of the semantics for textual explanations

4.2 Formal metrics for explanations

Metric gap

For each metric, we can evaluate the *metric gap* that will represent how far from an optimal MUS we can fall if we pick a random MUS from $\psi \in \mathcal{M}(H,Q)$. This notion has been introduced by Beynier *et al.* [2024] and has been adapted to our explanation framework.

Definition 4.3 (Metric Gap). *For a given metric m, the metric gap of m over $\mathcal{Q}_l$ with $l \in \{1,2\}$ is defined as the worst ratio over all negative instances between the worst value of m on a MUS and the best one, i.e.:*

$$MG(m,l) := \max_{\substack{H,\sigma\in\mathcal{H}\times\Sigma\\ Q\in\mathcal{Q}_l(H,\sigma)}} \max_{\psi,\psi^*\in\mathcal{M}(H,Q)} \frac{m_{H,Q}(\psi)}{m_{H,Q}(\psi^*)}$$

Explanations conciseness

We consider an instance $H = (\mathcal{A},\mathcal{O},\succ)$ and an allocation σ that satisfies r-EF. Let's consider questions Q^1_{loc} that might be asked by some agents.

Here is an example that we will use to illustrate some results:

Example 4.4. Consider an instance with n agents and objects where the preference are as is:

$$\begin{array}{rccccccccc}
1: & o_1 & \succ & o_2 & \succ & \dots & \succ & o_{n-1} & \succ & \boxed{o_n} \\
2: & \boxed{o_2} & \succ & o_3 & \succ & \dots & \succ & o_1 & \succ & o_n \\
3: & \boxed{o_3} & \succ & o_4 & \succ & \dots & \succ & o_2 & \succ & o_n \\
\dots & \dots & & \dots & & \dots & & \dots & & \dots \\
n-1: & \boxed{o_{n-1}} & \succ & o_1 & \succ & \dots & \succ & o_{n-2} & \succ & o_n \\
n: & \boxed{o_1} & \succ & o_2 & \succ & \dots & \succ & o_n & \succ & o_{n-1}
\end{array}$$

Consider the allocation $\sigma : i \mapsto \begin{cases} o_n & \text{if } i = 1 \\ o_1 & \text{if } i = n \\ o_i & \text{otherwise} \end{cases}$

This allocation is indeed r-EF.

Assuming that agent 1 asks the question $Q^1_{loc}(1,o_2)$ ("Why did I not received object o_2"), this problem is now UNSAT as it would violate the rank-envy-freeness. Indeed, based on the characterization, object o_2 must be affected to one of the agents who ranked it first if they exist, which is the case (agent 2).

Here are two possible MUSes we can extract of this setup with $n = 4$:

ψ^1	$\phi_Q = x_{1,o_2}$ $\phi_I^{topkr}(o_2) = x_{2,o_2}$ $\phi_{alloc}^{\leq 1,\mathcal{O}}(o_2, 1, 2) = \neg x_{1,o_2} \vee \neg x_{2,o_2}$
ψ^2	$\phi_Q = x_{1,o_2}$ $\phi_I^{topkr}(o_2) = x_{2,o_2}$ $\phi_I^{topkr}(o_3) = x_{2,o_3}$ $\phi_{alloc}^{\geq 1,\mathcal{O}}(o_1) = x_{1,o_1} \vee x_{2,o_1} \vee x_{3,o_1} \vee x_{4,o_1}$ $\phi_{alloc}^{\geq 1,\mathcal{O}}(o_4) = x_{1,o_4} \vee x_{2,o_4} \vee x_{3,o_4} \vee x_{4,o_4}$ $\phi_{alloc}^{\leq 1,\mathcal{A}}(1, o_1, o_2) = \neg x_{1,o_1} \vee \neg x_{1,o_2}$ $\phi_{alloc}^{\leq 1,\mathcal{A}}(1, o_2, o_4) = \neg x_{1,o_2} \vee \neg x_{1,o_4}$ $\phi_{alloc}^{\leq 1,\mathcal{A}}(2, o_1, o_2) = \neg x_{2,o_1} \vee \neg x_{2,o_2}$ $\phi_{alloc}^{\leq 1,\mathcal{A}}(2, o_2, o_4) = \neg x_{2,o_2} \vee \neg x_{2,o_4}$ $\phi_{alloc}^{\leq 1,\mathcal{A}}(3, o_1, o_3) = \neg x_{3,o_1} \vee \neg x_{3,o_3}$ $\phi_{alloc}^{\leq 1,\mathcal{A}}(3, o_3, o_4) = \neg x_{3,o_3} \vee \neg x_{3,o_4}$ $\phi_{alloc}^{\leq 1,\mathcal{A}}(4, o_1, o_4) = \neg x_{4,o_1} \vee \neg x_{4,o_4}$

Table 3: Muses obtained from example 4.4 with $n = 4$.

	#agents	#objects	#clauses	#variables	length	depth	breadth
ψ^1	2	1	3	2	4	4	1
ψ^2	4	4	12	11	49	4	16

Table 4: Metrics over the MUSes of the example 4.4

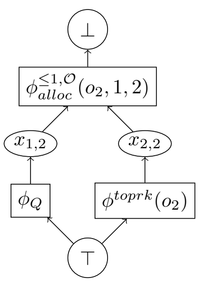

Figure 1: Satisfiability Implication Graph obtained from ψ^1

Proposition 4.5. *The tight lower and upper bounds for #agents (resp., #objects) involved in a MUS are* 2 *and* n *(resp.,* 1 *and* n*), respectively. The metric gap for #agents (resp., #objects) is thus* $\Theta(n)$*.*

Proposition 4.6. *The depth of an explanation is always 4.*

Proof. The construction of the satisfiability implication graph relies on 2 types of clauses: disjunctions of positive literals (at-least clauses, question clause, toprk and rk clauses), and disjunctions of negative literals (at-most clauses). All disjunctions of positive (*resp.* negative) literals will necessarily be connected to the node $\top$ (*resp.* $\bot$) and they will be separated by a layer of variable nodes, which makes a depth of 4. □

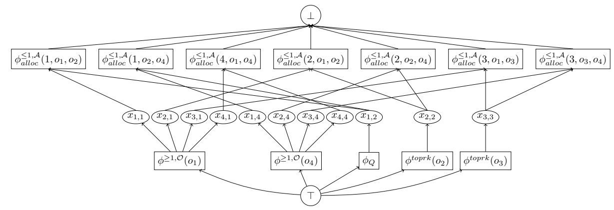

Figure 2: Satisfiability Implication Graph obtained from ψ^2

Proposition 4.7. *The tight lower bounds of #clauses, #variables, length and breadth are respectively 3, 2, 4 and 1.*

Proof. Tight lower bounds are obtained in the SIG of ψ^1. □

Data disclosure

Privacy disclosure occurs when an explanation allows the querying agent i^* to **eliminate preference profiles** that were previously consistent with her prior information. Here, the full matching is known to the agent.

Let $\mathcal{I}_{i^*}$ denote the prior information available to agent i^* before receiving any explanation. In the public allocation setting, $\mathcal{I}_{i^*}$ comprises the full allocation σ, the $\mathrm{r_k}$-EF axiom, and the agent's own preference order $\succ_{i^*}$.Let $\mathcal{P}$ denote a set of preference profiles.

Definition 4.8 (Prior compatible set)**.**

$$\Pi_{i^*}^{\text{prior}} \ := \ \{ \succ \in \mathcal{P} \mid \succ \ \textit{is consistent with } \mathcal{I}_{i^*} \}$$

That is, $\Pi_{i^*}^{\text{prior}}$ *collects all preference profiles over all agents that are jointly consistent with: (i) the known allocation* σ *satisfying* r_k*-EF , and (ii) the known preference order of* i^**.*

Definition 4.9 (Posterior compatible set)**.** *After receiving explanation Expl, agent* i^* *updates her beliefs by discarding all profiles incompatible with the explanation:*

$$\Pi_{i^*}^{\text{post}}(\textit{Expl}) \ := \ \left\{ \succ \in \Pi_{i^*}^{\text{prior}} \ \middle| \ \succ \ \textit{is consistent with Expl} \right\}$$

By construction, $\Pi_{i^*}^{\text{post}}(\textit{Expl}) \subseteq \Pi_{i^*}^{\text{prior}}$.

Definition 4.10 (Absolute Disclosure induced by the explanation)**.**

$$\Delta^{\text{abs}}(\textit{Expl}) \ := \ \left| \Pi_{i^*}^{\text{prior}} \right| - \left| \Pi_{i^*}^{\text{post}}(\textit{Expl}) \right|$$

Definition 4.11 (Relative disclosure)**.**

$$\Delta^{\text{rel}}(\textit{Expl}) \ := \ 1 - \frac{\left| \Pi_{i^*}^{\text{post}}(\textit{Expl}) \right|}{\left| \Pi_{i^*}^{\text{prior}} \right|}$$

$\Delta^{\text{rel}}(\textit{Expl}) = 0$ means the explanation discloses no information; $\Delta^{\text{rel}}(\textit{Expl}) = 1$ means the explanation fully determines the preference profile of other agents.

Let $\succ_{-j}$ be a preference profil for all agent but j.

Definition 4.12 (Marginal Compatible Set)**.** *We define the marginal compatible set of preferences for agent j:*

$$\Pi^{\text{prior}}_{i^*,j} \;:=\; \left\{ \succ_j \in \mathcal{P}_j \;\middle|\; \exists\ \succ_{-j} \in \prod_{\ell \neq j} \mathcal{P}_\ell : (\succ_{-j}, \succ_j) \in \Pi^{\text{prior}}_{i^*} \right\}$$

and analogously $\Pi^{\text{post}}_{i^*,j}(Expl)$.

Definition 4.13 (Agent-level disclosure)**.**

$$\Delta^{\text{rel}}_j(Expl) \;:=\; 1 - \frac{\left|\Pi^{\text{post}}_{i^*,j}(Expl)\right|}{\left|\Pi^{\text{prior}}_{i^*,j}\right|}$$

4.3 SAT formulation for data disclosure

We consider the Boolean variables $x_{i,j}$ to represent a matching σ i.e. $x_{i,j} = \top$ iff and the Boolean variables $p_{i,j,k}$ to represent a preference profile $\succ$ i.e. $p_{i,j,k} = \top$ iff $r_i(j) = k$. The constraints are adapted from the initial SAT formulation to apply to an unknown profile.

Formulation of the constraints for an unknown profile

The consitency constraints must be rephrased to take into account the unknown profiles.

- Profile consistency constraints:

$$\phi^{\text{atleast-rank}} := \bigwedge_{i,j \in [n]^2} \bigvee_{k \in [n]} p_{i,j,k}$$

$$\phi^{\text{atmost-rank}} := \bigwedge_{i,j \in [n]^2} \bigwedge_{\substack{k_1,k_2 \in [n]^2 \\ k_1 \neq k_2}} \neg p_{i,j,k_1} \vee \neg p_{i,j,k_2}$$

- Profile-independent r_k-EF : In this setup a direct, profile independent formulation of r_k-EF is used.

$$\phi^{\text{r}_\text{k}\text{-EF}} \;:=\; \bigwedge_{\substack{i,j \in \mathcal{A}^2 \\ i \neq j}} \phi^{r-k-EF}_{i,j}$$

and $\forall i,j \in \mathcal{A}^2, i \neq j$.

$$\phi^{\text{r}_\text{k}\text{-EF}}_{i,j} \;:=\; \bigwedge_{\substack{p,q \in \mathcal{O}^2 \\ p \neq q}} \bigwedge_{\substack{r,s \in [n]^2 \\ s<r}} \bigwedge_{\substack{t \in [n] \\ \min(k,s)<t}} \phi^{\text{no-envy}\binom{i,j,p,q,}{r,s,t}}$$

with $\forall i,j \in [n]^2, i \neq j, \forall p,q \in \mathcal{O}^2, p \neq q, \forall r,s \in [n]^2, s < r, \forall t \in [n], \min(k,s) < t$:

$$\phi^{\text{no-envy}\binom{i,j,p,q,}{r,s,t}} := \neg x_{i,p} \vee \neg x_{j,q} \vee \neg p_{i,p,r} \vee \neg p_{i,q,s} \vee \neg p_{j,q,t}$$

For readability, r, s, t are interpreted here as :

$r := r_i(p) \equiv$ the rank of object p by agent i

$s := r_i(q) \equiv$ the rank of object q by agent i

$t := r_j(q) \equiv$ the rank of object q by agent j

Expressing prior knowledge (disclosure induced by the matching itself)

The publicly known allocation σ already constrains the set of compatible preference profiles, independently of any explanation.

Lemma 4.14 (Allocation-induced constraints)**.** *Let σ be a public r_k-EF allocation with its rank known to the agents. For every pair of agents $i,j \in \mathcal{A}$, if $r_i(\sigma(j)) < r_i(\sigma(i))$ (i.e., agent i prefers $\sigma(j)$ to her own object), then agent i can deduce that $r_j(\sigma(j)) \leq \min\{r_i(\sigma(j)),\, k\}$*

Proof. Since the $x_{i,j}$ are known, the expression $\varphi^{\text{r}_\text{k}\text{-EF}}$ can be simplified into :

$$\varphi^{\text{r}_\text{k}\text{-EF}} \;\equiv\; \bigwedge_{\substack{i,j \in \mathcal{A}^2 \\ i \neq j}} \bigwedge_{\substack{r,s \in [n]^2 \\ s<r}} \bigwedge_{\substack{t \in [n] \\ \min(k,s)<t}} \neg p_{i,\sigma(i)r} \vee \neg p_{i,\sigma(j),s} \vee \neg p_{j,\sigma(j),t}$$

Let $(i,j) \in \mathcal{A}^2, (r,s) \in [n]^2$ such that $i \neq j$ and $s < r$: Applying the consistency constraints, we have :

$$\begin{aligned}
&\bigwedge_{\substack{t \in [n] \\ \min(k,s)<t}} \neg p_{i,\sigma(i),r} \vee \neg p_{i,\sigma(j),s} \vee \neg p_{j,\sigma(j),t} \\
&\equiv \neg \bigvee_{\substack{t \in [n] \\ \min(k,s)<t}} p_{i,\sigma(i)r} \wedge p_{i,\sigma(j),s} \wedge p_{j,\sigma(j),t} \\
&\equiv \forall t > \min(k,s), (p_{i,\sigma(i),r} \wedge p_{i,\sigma(j),s} \Rightarrow \neg p_{j,\sigma(j),t}) \\
&\equiv (p_{i,\sigma(i),r} \wedge p_{i,\sigma(j),s}) \Rightarrow (\forall t > \min(k,s), \neg p_{j,\sigma(j),t}) \\
\text{cons.} \to &\equiv (p_{i,\sigma(i),r} \wedge p_{i,\sigma(j),s}) \Rightarrow (\exists! t \leq \min(k,s), p_{j,\sigma(j),t}) \\
&\equiv (r_i(\sigma(i)) = r \wedge r_i(\sigma(j)) = s) \\
&\qquad \Rightarrow r_j(\sigma(j)) <= \min(k,s)
\end{aligned}$$

We can deduce that $\forall (i,j) \in \mathcal{A}^2, (r,s), r_i(\sigma(j)) < r_i(\sigma(i)) \Rightarrow r_j(\sigma(j)) \leq \min(r_i(\sigma(j)), k)$

□

The constraints derivable from the allocation alone are compiled into:

$$\Phi^{\text{alloc}} \;:=\; \bigwedge_{\substack{i,j \in \mathcal{A} \\ r_i(\sigma(j))<r_i(\sigma(i))}} r_j(\sigma(j)) \leq \min\{r_i(\sigma(j)),\, k\}$$

This formula characterizes the prior information that i^* can derive from σ alone, independently of any MUS-based explanation.

Expressing posterior knowledge after an explanation (disclosure induced by the r_k-EF explanation)

As the R_kEF clauses are the only one depending on the preference profile, they are the only clauses that imply constraints on the possible profiles.

Lemma 4.15 (Encoding condition for the direct formulation)**.** *The clause $\neg x_{i,p} \vee \neg x_{j,q}$ is included in the SAT encoding of R_kEF_{direct} if and only if: $r_i(q) < r_i(p)$ and $\big(r_i(q) < r_j(q) \,\vee\, r_j(q) > k\big)$.*

Proof. Direct substitution into the definition of the rank-k-envy-freeness.

This lemma directly entails :

Lemma 4.16 (Disclosed constraints from a single clause)**.** *If the clause $\neg x_{i,p} \vee \neg x_{j,q}$ appears in the MUS returned to agent i^*, then i^* can deduce:*

$$r_i(q) < r_i(p) \quad \text{and} \quad \big(r_i(q) < r_j(q) \ \vee\ r_j(q) > k\big)$$

The natural-language reading of this disclosure is *"Agent i prefers object q over object p, and either agent j ranks q worse than agent i does, or agent j ranks q outside her top-k."*

Definition 4.17 (Explanation disclosure formula)**.** *Given an explanation Expl consisting of r_k-EF clause instances, the disclosure formula is:*

$$\Phi^{r_k\text{-}EF}(Expl) := \bigwedge_{\substack{(i,j,p,q) \\ r_k\text{-}EF(i,j,p,q) \in Expl}} \begin{bmatrix} (r_i(q) < r_i(p) \wedge r_i(q) < r_j(q)) \\ \vee\ r_j(q) > k \end{bmatrix} \tag{1}$$

Proposition 4.18 (Per-clause monotonicity in k)**.** *Fix a clause $r_k EF(i,j,p,q)$ present in the encodings of both thresholds k and k' with $k' > k$.* ***The constraint disclosed by this clause at threshold k' is at least as strong as at threshold k :*** *every preference profile satisfying the k'-disclosure also satisfies the k-disclosure.*

Proof. The disclosed constraints at threshold k are:

$$C_1 : \quad r_i(q) < r_i(p)$$

$$C_2^k : \quad r_i(q) < r_j(q) \ \vee\ r_j(q) > k$$

and at threshold k':

$$C_2^{k'} : \quad r_i(q) < r_j(q) \ \vee\ r_j(q) > k'$$

Since $k' > k$, we have $r_j(q) > k' \implies r_j(q) > k$. Therefore, for any fixed values of $r_i(q)$ and $r_j(q) : C_2^{k'} \implies C_2^k$. The constraint C_1 is identical at both thresholds. Hence any profile satisfying $C_1 \wedge C_2^{k'}$ also satisfies $C_1 \wedge C_2^k$, but not conversely. The disclosure at threshold k' is therefore at least as strong. □

Definition 4.19 (Agent footprint of an explanation)**.** *The agent footprint of explanation Expl is the set of agents appearing in at least one clause of the MUS:*

$$\mathcal{A}(Expl) := \left\{ \begin{array}{l|l} a \in \mathcal{A} & \exists (i,j,p,q) \\ & r_k\text{-}EF(i,j,p,q) \in Expl, \\ & a \in \{i,j\} \end{array} \right\} \tag{2}$$

Note that if $j \notin \mathcal{A}(Expl)$, then $\Delta_j^{\text{rel}}(Expl) = 0$: the explanation discloses no information about agent j's preferences.

5 Algorithmic Contributions

5.1 From an Instance H, an allocation σ and a Question q to a MUS

Given an instance H, an allocation σ and a question q of type 1 or 2, this subsection describes the process of extracting a Minimally Unsatisfiable Subset (MUS) from H, σ and q.

Proposition 5.1 (Feasibility for Type 2 Questions)**.** *If we only encode the CNF of the allocation problem and a question q of type 2, the resulting CNF will always be satisfiable.*

Proof. Consider an agent i^* who asks a question of type Q^2_{loc}. Let o be the object ranked first by i^*, i.e., $r_{i^*}(o) = 1$. By applying algorithm 1, we can modify the candidate selection at the first step to pick the pair (i^*, o), as this pair has minimal rank. Continuing the algorithm as specified will then produce a complete rank-envy-free allocation σ'.

Since the problem is satisfiable, no MUS can be extracted for a question of type Q^2_{loc} when degradation is not prevented for agents. Indeed, the agent could receive a better object than the one in their current allocation σ, specifically their top-ranked object o. □

Corollary 5.2. *From the proof, we can also deduce that Q^1_{loc} where the asked object is ranked first will also be satisfiable.*

Here is the algorithm we suggested to extract a MUS.

Algorithm 2: Algorithm for Extracting a MUS from H and q

```
Let H = (A, O, ≻)
Let q be a question of type 1 or 2
Let σ be the current allocation
Encode the CNF for the allocation problem H
Add the clause corresponding to q as a desired
  property
if q is of type 1 then
    Attempt to extract a MUS from the CNF
    if A MUS is found then
        return the MUS
if No MUS is found or q is of type 2 then
    encode a non-degradation condition based on σ
    return the MUS derived from the non-degradation
      condition
```

Proposition 5.3 (Guaranteed Return of a MUS)**.** *Algorithm 2 will necessarily return a MUS.*

Proof. The algorithm first attempts to extract a MUS directly for questions of type 1. If this fails or for questions of type 2, it encodes a non-degradation condition based on the current allocation σ. Since any rank-envy-free allocation is Pareto optimal, any alternative rank-envy-free allocation would degrade at least one agent's allocated item. Thus, the non-degradation condition ensures the existence of a MUS, guaranteeing that the algorithm will always return a result. □

5.2 Assessing Experimentally the Preference Disclosure Induced by MUS Explanations

Definition 5.4 (Most dissatisfied agent)**.** *Agent i^* is the most dissatified agent iff* $r_{i^*}(\sigma(i^*)) = \max_{j\in\mathcal{A}}(r_j(\sigma(j)))$

Motivation. Providing an agent with a MUS explanation is not epistemically neutral: the clauses of the MUS encode structural constraints on the allocation problem that may allow the recipient to narrow down the set of preference profiles consistent with the observed outcome. We formalise and quantify this *preference disclosure* effect through an epistemic framework grounded in profile enumeration.

Epistemic framework. Let i^* be the most dissatisfied agent (Definition 5.4). We model her knowledge as the set of profile–allocation pairs that are consistent with what she observes. Formally, i^* knows her own preference order π_{i^*}, the full matching μ, and the consistency and $\mathrm{r_k}$-EF axioms; she does *not* know the preference orders of other agents.

Experimental pipeline. Algorithm 3 (RUNSINGLESEED) implements the complete per-instance measurement pipeline. Preference profiles are sampled from the Mallows model Mallows [1957] with dispersion parameter $\varphi \in (0, 1]$: $\varphi \to 0$ concentrates mass on the reference ranking, while $\varphi = 1$ yields the uniform distribution.

The procedure is as follows:

1. A profile $\mathcal{P}$ is sampled and an allocation σ satisfying $\mathrm{r_k}$-EF is computed; instances for which the SAT formula is unsatisfiable are discarded.
2. The most dissatisfied agent i^* is identified and asks for her top-ranked object $o^* = \pi_{i^*}(1)$; if the augmented formula is satisfiable, no MUS exists and the instance is labelled NOMUS.
3. Otherwise, a MUS M is extracted and the epistemic formula $\mathcal{F}_{\mathrm{bt}}$ is constructed.
4. Π_{prior} and Π_{post} are enumerated by model counting over $\mathcal{F}_{\mathrm{bt}}$ before and after injecting the explanation constraints; the disclosure metrics are then computed.

Correctness guarantee. The enumeration of Π_{prior} is *complete*: every profile–allocation pair consistent with i^*'s prior knowledge is generated. This completeness underpins the validity of the disclosure measurements; any incompleteness would lead to underestimating Δ_{rel}.

6 Experiments and Results

6.1 Experimental setting

The goal of the experiments is to assess the data disclosure and the explanation conciseness in different instance $H = (\mathcal{A}, \mathcal{O}, \succ)$.

The profiles are obtained thanks to the Mallows [1957] model with several values for the parameter φ representing heterogeneity of the preferences of the agents (here, 0.25, 0.375, 0.5, 0.625 and 0.75).

Two different setups are considered, one for the conciseness and clarity study and another one for the data disclosure study. In a later work, the framework will be uniformed in order to compare the two metrics type.

Conciseness setup

For the explanation's conciseness, we consider instances H with allocations σ that are r-EF. The axioms to grant a valid allocation are encoded with redundancy, and the r-EF is expressed under its characterization form. is considered as well as the consistency axiom (exactly one object per agent and each object is matched to exactly one agent). Since through the serial dictatorship algorithm, there is always a matching for a given instance that allows agent i^* to be matched with their favourite resource. Therefore, we only consider agents i^* who got an object with a rank s.t. $r_{i^*} > 2$ and who asks for an object q^* ranked second or worse. For the questions that lead to an UNSAT problem, we extract the shortest MUS in number of clauses and compute the metrics of conciseness.

Data disclosure setup

For the data disclosure the $\mathrm{r_k}$-EF and its direct formulation is considered. The agent querying the counterfactual question is the most dissatisfied agent.Please note that sometimes no MUS and therefore no explanation is possible since the agent i^* could have gotten is favourite object. Data disclosure is only measured in the cases where an explanation is possible.

6.2 Results on clarity and conciseness

A critical aspect of generating explanations is ensuring they are computationally tractable and concise for practical usability. To assess this, we conducted experiments across 200 random seeds for varying numbers of agents, $n \in \{3, \dots, 8\}$.

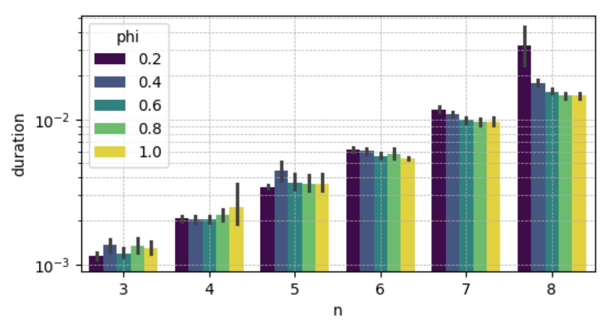

Figure 4: Duration to extract a MUS for a question Q^1_{loc} with regards to n

Our results, illustrated in Figure 4, demonstrate that the time required to extract a minimal-size MUS grows exponentially with the number of agents n. Due to computational constraints, we limited our analysis to $n \leq 8$, as preliminary tests with $n = 9$ indicated prohibitively long runtimes for some instances.

Algorithm 3: RunSingleSeed

Input : $N \in \mathbb{N}$, $k \in \{1, \ldots, N-2\}$, $\varphi \in (0, 1]$, $seed \in \mathbb{N}$
Output: $\Delta^{(\sigma)}_{\text{abs}}$, $\Delta^{(\sigma)}_{\text{rel}}$, $\big(\Delta^{(\sigma)}_{\text{rel},j}\big)_{j\in[N]}$

▷ **Step 1**: Profile generation
$\mathcal{P} \leftarrow \textsc{Mallows}(N, \varphi, \sigma)$
▷ **Step 2**: Phase-1 solver under r_k-EF
$\mathcal{F} \leftarrow \textsc{ConsistencyAxioms}(N) \wedge \text{r}_\text{k}\text{-EF}(\mathcal{P})$
$\mu \leftarrow \textsc{SatSolve}(\mathcal{F})$
if $\mu = \perp$ **then return** AllocFailure
▷ **Step 3**: Identification of the most dissatisfied agent
i^* $i^* \leftarrow \arg\max_{i\in[N]} \text{rank}_{\pi_i}(\mu(i))$
$q^* \leftarrow \pi_{i^*}(1)$ `// top-ranked object of` i^*
▷ **Step 4**: Agent query (*AgentQuestion*)
$\mathcal{F}' \leftarrow \mathcal{F} \wedge \big(x_{i^*, o^*} = 1\big)$
$\mu' \leftarrow \textsc{SatSolve}(\mathcal{F}')$
if $\mu' \neq \perp$ **then return** NoMus ▷ query satisfiable; no explanation
▷ **Step 5**: MUS extraction ($\mathcal{F}'$ is UNSAT)
$M \leftarrow \text{MUS}(\mathcal{F}')$
// $M \subseteq \text{clauses}(\mathcal{F}')$,
M *is UNSAT,*
$\forall C \in M :\ M \setminus \{C\}$ *is SAT*
▷ **Step 6**: Backtrack solver (epistemic view of i^*)
$\mathcal{P}_\emptyset \leftarrow \textsc{EmptyProfile}(N)$
▷ Other agents preferences unknown
$\mathcal{F}_{\text{bt}} \leftarrow \textsc{ConsistencyAxioms}(N)$
$\wedge\ \textsc{KnownPartialPreference}(i^*, \pi_{i^*})$
$\wedge\ \textsc{KnownMatching}(\mu)$
$\wedge\ \textsc{RkEF-EveryProfile}(k, \mathcal{P}_\emptyset)$
assert $\textsc{SatSolve}(\mathcal{F}_{\text{bt}}) \neq \perp$

$\wedge\ \textsc{KnownPartialPreference}(i^*, \pi_{i^*})$
$\wedge\ \textsc{KnownMatching}(\mu)$
$\wedge\ \textsc{RkEF-EveryProfile}(k, \mathcal{P}_\emptyset)$
assert $\textsc{SatSolve}(\mathcal{F}_{\text{bt}}) \neq \perp$
▷ **Step 7a** : Enumeration of Π_{prior}
$\Pi_{\text{prior}} \leftarrow \{\, (\mathcal{P}', \mu') \mid (\mathcal{P}', \mu') \models \mathcal{F}_{\text{bt}} \,\}$
for $j \in [N]$ **do**
  $\Pi^{\text{prior}}_j \leftarrow \{\, \pi'_j \mid \exists\, \mu' : (\mathcal{P}', \mu') \in \Pi_{\text{prior}} \,\}$
end
▷ Step 7b: Injection of MUS explanation constraints (posterior) $\mathcal{F}_{\text{bt}} \leftarrow \mathcal{F}_{\text{bt}} \wedge \textsc{ExplanationConstraints}(M, \mu)$
▷ **Step 7c**: Enumeration of Π_{post}
$\Pi_{\text{post}} \leftarrow \{\, (\mathcal{P}', \mu') \mid (\mathcal{P}', \mu') \models \mathcal{F}_{\text{bt}} \,\}$
for $j \in [N]$ **do**
  $\Pi^{\text{post}}_j \leftarrow \{\, \pi'_j \mid \exists\, \mu' : (\mathcal{P}', \mu') \in \Pi_{\text{post}} \,\}$
end
▷ **Step 8**: Disclosure metrics
$\Delta^{(\sigma)}_{\text{abs}} \leftarrow |\Pi_{\text{prior}}| - |\Pi_{\text{post}}|$

$$\Delta^{(\sigma)}_{\text{rel}} \leftarrow \begin{cases} 1 - \dfrac{|\Pi_{\text{post}}|}{|\Pi_{\text{prior}}|} & \text{if } |\Pi_{\text{prior}}| > 0, \\ 0 & \text{otherwise.} \end{cases}$$

for $j \in [N]$ **do**

$$\Delta^{(\sigma)}_{\text{rel},j} \leftarrow \begin{cases} 1 - \dfrac{|\Pi^{\text{post}}_j|}{|\Pi^{\text{prior}}_j|} & \text{if } |\Pi^{\text{prior}}_j| > 0, \\ 0 & \text{otherwise.} \end{cases}$$

end
return $\Delta^{(\sigma)}_{\text{abs}}$, $\Delta^{(\sigma)}_{\text{rel}}$, $\big(\Delta^{(\sigma)}_{\text{rel},j}\big)_{j\in[N]}$

Figure 3: Algorithm RunSingleSeed(N, k, φ, σ) — Disclosure measurement for one instance.

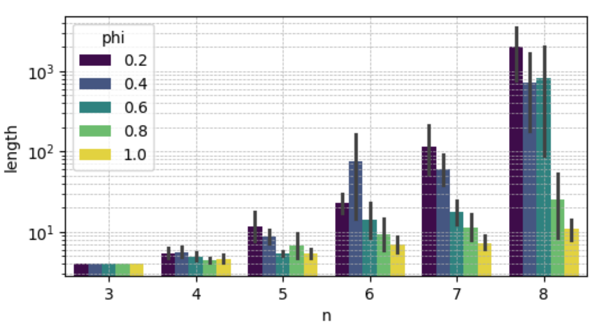

Figure 5: Length of the answers for a question Q^1_{loc} with regards to n

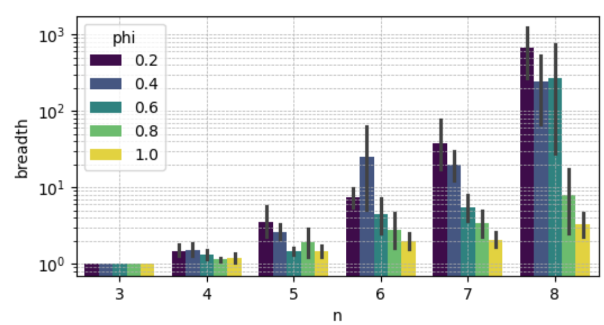

Figure 6: Breadth of the answers for a question Q^1_{loc} with regards to n

Beyond computational efficiency, the clarity and conciseness of explanations are essential for user comprehension. However, our findings reveal that both the length and breadth of explanations increase rapidly with n. As shown in Figures 5 and 6, the length of explanations (number of clauses) and their breadth (diversity of clauses) exhibit combinatorial growth as n increases. This trend suggests that explanations quickly become too lengthy and complex for users to interpret effectively, undermining their practical utility.

Another interesting result is that the more agent's preferences differs, *i.e.* $\varphi \approx 1$, the shorter the explanations. We can explain this because questions Q^1_{loc} can be closed faster with ϕ_H^{top}, and with highly randomized profiles, more objects are in T, thus less clauses and variables are in the SIG.

6.3 Results on Preference Disclosure

Due to the computational cost of full profile enumeration, the disclosure analysis is restricted to $N \in \{3, 4\}$ agents. For each tuple (N, k, φ), we sample 10,000 seeds. We restrict attention to $k < N - 1$: for $k \geq N - 1$ the $RkEF(k)$ constraint becomes equivalent to serial dictatorship, and the agent's query is satisfiable, producing no MUS and hence no disclosure event worth measuring. The number of MUS instances n_{mus} per configuration is reported in Table 5.

A first transversal observation is that $\bar{\Delta}_{\text{rel}}$ is remarkably stable across all values of φ within each (N, k) configuration (Table 5, column std for Δ_{rel}). Preference disclosure is therefore a structural property of the pair (N, k), not of the preference distribution. We now examine each configuration in turn.

The degenerate case $N = 3$, $k = 1$: exact combinatorial disclosure. For $(N = 3,\ k = 1)$, the empirical distribution of Δ_{rel} is degenerate at the single value $22/23 = 0.\overline{954545}$ across all seeds and all values of φ (std $\approx 2 \times 10^{-16}$, i.e. floating-point precision; Shapiro–Wilk $W = 1, p = 1$). This is not a statistical regularity but an exact algebraic identity: $|\Pi_{\text{prior}}| = 23$ and $|\Pi_{\text{post}}| = 1$ deterministically. The MUS reduces the space of compatible profiles to a *unique* profile, i.e. the preferences of all other agents are fully identified.

Proposition 6.1 (Exact prior size for $N = 3,\ k = 1$)**.** *For $N = 3$ and $k = 1$, we have $|\Pi_{\text{prior}}| = 23$ and $|\Pi_{\text{post}}| = 1$, hence $\Delta_{\text{rel}} = 22/23$ deterministically.*

The effect of φ is confined to the *frequency* of MUS instances: n_{mus} decreases from 726 ($\varphi = 0.25$) to 438 ($\varphi = 0.75$). As preferences become more dispersed, the most dissatisfied agent is less likely to face a structurally impossible query, but when she does, the disclosure is invariably total in the case of $N = 3, k = 1$.

Bimodal disclosure for $N = 4$, $k = 1$. For $(N = 4,\ k = 1)$, the distribution of Δ_{rel} is no longer degenerate but remains highly concentrated near 1 ($\bar{\Delta}_{\text{rel}} \in [0.977, 0.980]$, std ≈ 0.016). The Shapiro–Wilk test rejects normality strongly ($p < 10^{-34}$), with negative skewness (-0.97 to -1.19) and kurtosis near -2 at large φ, consistent with a bimodal distribution.

The strip plots of $|\Pi_{\text{prior}}|$ (Figure 8) reveal the source of this bimodality: $|\Pi_{\text{prior}}|$ takes values in two well-separated clusters, a low cluster around 1,184–1,240 and a high cluster around 4,218. These correspond to two structurally distinct classes of MUS instances; the nature of this distinction — whether it reflects the depth of the RkEF conflict, the size of

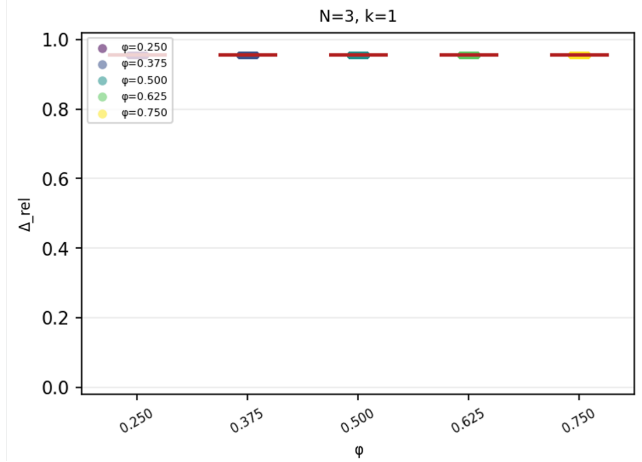

Figure 7: Δ_{rel} vs. φ for $N = 3$, $k = 1$ (10 000 seeds per point). The distribution is degenerate at $21/22$ for all φ.

the MUS, or another combinatorial property — remains an open question (see Section 6.3).

Despite this bimodality in $|\Pi_{\text{prior}}|$, the relative disclosure Δ_{rel} remains stable around 0.978. Crucially, $\bar{\Delta}_{\text{abs}}$ grows with φ (from 1,855 to 2,400, Table 5), while $\bar{\Delta}_{\text{rel}}$ remains nearly constant. This implies that $|\Pi_{\text{post}}|$ grows proportionally with $|\Pi_{\text{prior}}|$ as φ increases: the MUS eliminates a constant *fraction* of the compatible space, regardless of the absolute size of that space. This is a structural regularity that does not follow trivially from the definition of the MUS.

Quasi-certain near-total disclosure for $N = 4$, $k = 2$. For $(N = 4,\ k = 2)$, the disclosure is even stronger: $\bar{\Delta}_{\text{rel}} \approx 0.9995$ with std ≈ 0.0003, meaning that $|\Pi_{\text{post}}|$ is reduced to a handful of compatible profiles. Inspection of $|\Pi_{\text{post}}|$ (Figure 10) reveals that it takes only a small number of discrete integer values in $\{2, 3, 4, 6, 8\}$, reflecting a discrete combinatorial structure of the posterior space. This discreteness directly produces the step-like behaviour observed in Δ_{rel} as a function of φ.

Meanwhile, $\bar{|}\Delta_{abs}|$ concentrates near 6,700, which is much larger than in the $k = 1$ case ($\sim 1854 - 2400$). A similar observation can be made for $\bar{|}\Delta_{rel}|$. This apparent paradox i.e. a more permissive axiomatic constraint (R_2EF) yields more absolute and relative data disclosure than a more restrictive one ($R1EF$), is consistent by proposition 4.18. A more general proposition is yet an open question : does a higher k always lead to a higher data disclosure ?

The non-monotone behaviour of $\bar{\Delta}_{\text{rel}}$ at $\varphi = 0.625$ (visible in Figure 11) should not be over-interpreted: with only $n_{\text{mus}} = 83$ and 54 instances at $\varphi \in \{0.625, 0.750\}$, the confidence intervals are wide and the observed peak falls within statistical noise. Estimates for $(N = 4,\ k = 2,\ \varphi \geq 0.625)$ should be treated with caution.

Per-agent disclosure and fairness implications. The aggregate Δ_{rel} conceals heterogeneity across agents. Figure 12 show the mean per-agent relative disclosure $\bar{\Delta}_{\text{rel},j}$ for $N = 3$ and $N = 4$. In both cases, disclosure is concentrated on the agents who received the *worst* allocations, i.e. those with

Table 5: Summary statistics of Δ_{rel} and Δ_{abs} per configuration (N, k, φ). Only MUS instances are included. CI_{95} denotes the 95 % confidence interval for the mean. W is the Shapiro–Wilk statistic ($p < 0.05$ rejects normality).

N	k	φ	n_{mus}	$\bar{\Delta}_{\mathrm{rel}}$	med.	std	CI_{95} lo	CI_{95} hi	$\bar{\Delta}_{\mathrm{abs}}$	std
3	1	0.250	726	0.9545	0.9545	≈ 0	0.9545	0.9545	21.0	0
3	1	0.375	622	0.9545	0.9545	≈ 0	0.9545	0.9545	21.0	0
3	1	0.500	520	0.9545	0.9545	≈ 0	0.9545	0.9545	21.0	0
3	1	0.625	432	0.9545	0.9545	≈ 0	0.9545	0.9545	21.0	0
3	1	0.750	438	0.9545	0.9545	≈ 0	0.9545	0.9545	21.0	0
4	1	0.250	1402	0.9774	0.9809	0.0164	0.9765	0.9783	1854.7	1218.2
4	1	0.375	1626	0.9773	0.9809	0.0168	0.9765	0.9781	2048.5	1333.7
4	1	0.500	1522	0.9787	0.9809	0.0163	0.9779	0.9795	2142.9	1374.8
4	1	0.625	1357	0.9798	0.9830	0.0159	0.9790	0.9806	2345.7	1440.6
4	1	0.750	1224	0.9803	0.9830	0.0155	0.9794	0.9812	2399.7	1452.0
4	2	0.250	509	0.9994	0.9996	0.00035	0.99937	0.99944	6692.0	2.4
4	2	0.375	309	0.9995	0.9996	0.00030	0.99942	0.99949	6692.4	2.0
4	2	0.500	158	0.9995	0.9996	0.00026	0.99944	0.99952	6692.5	1.8
4	2	0.625	83	0.9995	0.9996	0.00024	0.99946	0.99957	6692.8	1.6
4	2	0.750	54	0.9995	0.9996	0.00022	0.99943	0.99954	6692.6	1.5

high allocated rank. For $N = 3$, $k = 1$: Agent 3 (77%) and Agent 2 (72%) are more exposed than Agent 1 (9%). For $N = 4$, $k = 2$: Agent 4 reaches 92% disclosure, while Agent 1 is at 35%. This gradient suggests that the MUS systematically leaks information about those agents whose preferences constrain the most dissatisfied agent's outcome, which are precisely the agents holding the objects she desires.

The no-MUS rate and its interaction with (k, φ). Table 5 shows that n_{mus} decreases both with k (for fixed N) and with φ (for fixed N, k), and these two effects compound: at $(N = 4, k = 2, \varphi = 0.75)$ only 54 MUS instances are observed out of 10,000. The decrease with k reflects the increased flexibility of a more permissive fairness constraint: under $R_kEF(2)$, it is less frequently the case that the most dissatisfied agent cannot receive a better object. The decrease with φ reflects the reduced competition for top-ranked objects when preferences are more dispersed. These two effects are empirically separable and roughly additive.

These two observations jointly produce a non-trivial policy tension. On the one hand, increasing k reduces the frequency of explanation possibility for $Q_{1,loc}$ and therefore reduces disclosure events (n_{mus} falls sharply). On the other hand, when a MUS does occur under a higher k, the resulting disclosure is strictly greater, both in absolute terms (Δ_{abs}) and in relative terms (Δ_{rel}), than under a lower k. A system designer cannot therefore avoid disclosure by relaxing the fairness constraint: doing so makes explanations harder to obtain but, conditionally on their occurrence, more severe.

Open questions. Several questions remain open and constitute directions for future work.

1. **Analytical derivation of $|\Pi_{\mathrm{prior}}|$ and $|\Pi_{\mathrm{post}}|$.** The exact values $|\Pi_{\mathrm{prior}}| = 22$ and $|\Pi_{\mathrm{post}}| = 1$ for $(N = 3, k = 1)$ are empirically certain but the combinatorial proof is incomplete (Proposition 6.1). More generally, establishing closed-form expressions for $|\Pi_{\mathrm{prior}}|$ and $|\Pi_{\mathrm{post}}|$ as functions of (N, k) would fully characterise the worst-case disclosure.
2. **Bimodality of $|\Pi_{\mathrm{prior}}|$ for $(N = 4, k = 1)$.** The two clusters at $\approx$ 1,200 and $\approx$ 4,200 suggest two structurally distinct classes of MUS instances. Identifying the combinatorial property that separates them (MUS size, number of non-degradation clauses, depth of the RkEF conflict, etc.) would clarify *why* disclosure is uniformly high across both populations.
3. **Scalability.** All results are limited to $N \leq 4$ due to the exponential cost of exact profile enumeration. Extending the analysis to larger instances requires either approximate counting methods (e.g. MCMC-based model counting) or structural results that bound Δ_{rel} without full enumeration.
4. **Disclosure under alternative formualtions of the REF.** The direct formulation of the $RkEF$ is one specific formulation of this axiom. Restrictive and characterization formulation (see Ouerdane *et al.* [2025] for the definition of those formulations under REF axiom) could also be explored and compared to the direct formulation in terms of data disclosure.

7 Conclusion and Future Works

In this paper, we studied the tension between explainability and preference privacy in house allocation problems. We proposed a SAT-based framework in which local counterfactual queries are answered by extracting Minimal Unsatisfiable Subsets (MUSes) from an encoding of the allocation, fairness axioms, and agent preferences under rank-k-envy-freeness ($\mathrm{r_k}$-EF). We introduced quantitative metrics for both explanation conciseness and preference disclosure, and established a monotonicity result showing that more permissive fairness constraints yield strictly stronger per-clause disclosure. Experimentally, we found that disclosure is a structural property of (N, k), largely independent of the preference distribution, and that MUS-based explanations expose almost all compatible preference profiles, with Δ_{rel} consistently above 0.95 across all tested configurations. Disclosure is furthermore heterogeneous: agents holding the objects most desired by the querying agent are the most exposed. These findings reveal a non-trivial policy tension, relaxing the fairness constraint reduces the frequency of disclosure events, but makes each occurrence strictly more severe.

Future works. Several directions remain open. From a theoretical standpoint, deriving closed-form expressions for $|\Pi_{\text{prior}}|$ and $|\Pi_{\text{post}}|$ as functions of (N, k) would fully characterise worst-case disclosure; in particular, we were not able to complete the combinatorial proof of Proposition 6.1 within the scope of this project. Comparing disclosure across the direct, characterisation, and restriction formulations of $\mathrm{r_k}$-EF is another direction we would have liked to explore. On the algorithmic side, scaling the disclosure pipeline beyond $N = 4$ via approximate model counting, and designing MUS selection strategies that explicitly navigate the conciseness–privacy trade-off, are the most immediate open problems. Finally, extending the framework to type-2 local questions (Q^2_{loc}) and to interactive, multi-round explanation protocols would broaden its practical applicability.

References

Tim Miller. Explanation in artificial intelligence: Insights from the social sciences. *Artificial Intelligence*, 267:1–38, 2019.

Peter Lipton. Contrastive Explanation. *Royal Institute of Philosophy Supplement*, 27:247–266, 1990.

Alejandro Arrieta, Natalia Díaz-Rodríguez, Javier Del Ser, Aditya K. Ghose, and Timothy M. Hospedales. Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges toward Responsible AI. *Information Fusion*, 58, 2020.

Felix Brandt, Vincent Conitzer, Ulle Endriss, Jérôme Lang, and Ariel D. Procaccia. *Handbook of Computational Social Choice*. Cambridge University Press, 2016.

Khaled Belahcene, Yann Chevaleyre, Christophe Labreuche, Nicolas Maudet, Vincent Mousseau, and Wassila Ouerdane. Accountable Approval Sorting. In *Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence (IJCAI)*, pages 70–76, 2018.

Khaled Belahcène, Vincent Mousseau, and Anaëlle Wilczynski. Combining Fairness and Optimality when Selecting and Allocating Projects. In *Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence (IJCAI-21)*, pages 38–44. International Joint Conferences on Artificial Intelligence Organization, 2021.

Alec Beynier, Yann Chevaleyre, Laurent Gourvès, Julien Lesca, Nicolas Maudet, and Piotr Wilczynski. Local Envy-Freeness in House Allocation Problems. *Journal of Autonomous Agents and Multi-Agent Systems*, 33(5), 2019.

Alec Beynier, Théo Mailly, Nicolas Maudet, and Piotr Wilczynski. Explaining the Lack of Locally Envy-Free Allocations. In *Proceedings of the 27th European Conference on Artificial Intelligence (ECAI)*, 2024.

Amandine Boixel and Ulle Endriss. Automated Justification of Collective Decisions via Constraint Solving. In *Proceedings of the 19th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2020.

David Gunning, Mark H. Stefik, et al. XAI—Explainable Artificial Intelligence. *Science Robotics*, 4(37), 2019.

Loustalot Knapp. Justification of Matching Outcomes. Master of Logic thesis, ILLC Amsterdam, 2022.

Mark H. Liffiton and Karem A. Sakallah. Algorithms for Computing Minimal Unsatisfiable Subsets of Constraints. *Journal of Automated Reasoning*, 40(1), 2008.

Mark H. Liffiton, Alessandro Previti, Sharad Malik, and Joao Marques-Silva. Fast, Flexible MUS Enumeration. *Constraints*, 21(2), 2016.

C. L. Mallows. Non-null ranking models. I. *Biometrika*, 44(1–2):114–130, 1957.

Alessandro Nardi, Amandine Boixel, and Ulle Endriss. A Graph-Based Algorithm for the Automated Justification of Collective Decisions. In *Proceedings of the 21st International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2022.

W. Ouerdane, et al. Explaining Why Fair Roommate Matchings Do Not Exist. 2025.

Rasoul Ramezanian and Mehdi Feizi. Stepwise ordinal efficiency for the random assignment problem. *Journal of Mathematical Economics*, 2020.

Fuhito Kojima and M. Utku Ünver. The "Boston" school-choice mechanism: an axiomatic approach. *Economic Theory*, 55(3):515–544, 2014.

Sina Mohseni, Niloofar Zarei, and Eric D. Ragan. A multidisciplinary survey and framework for design and evaluation of explainable AI systems. *ACM Transactions on Interactive Intelligent Systems*, 11(3–4):1–45, 2021.

Alexey Ignatiev, Nina Narodytska, Nicholas Asher, and Joao Marques-Silva. From contrastive to abductive explanations and back again. In *International Conference of the Italian Association for Artificial Intelligence (AIxIA 2020)*, pages 335–355, 2020. doi:10.1007/978-3-030-77091-4-21.

8 Appendix

8.1 Implementation details

The implementation translates the theoretical axiom framework into a practical SAT-based system for fair allocation and explainability. The codebase is organized in a python module splitting allocation modeling, axiom specification, and explanation generation.

Core Architecture: Models and Variables

The implementation grounds itself in the formal model through the `OrdinalPreferenceProfile` class, which encapsulates agent preferences as ordinal rankings over items. The matching itself is represented by the `Allocation` class, storing a bijective mapping between agents and resources.

Central to the SAT-based approach is the `VariablePool`, which manages the propositional variables underlying the CNF formulas. In particular, the pool maintains the boolean variables $x_{i,j}$ representing the assignment of agent i to resource j.

Axiom Abstraction and CNF Generation

The theoretical axioms described in the mathematical framework are implemented through an abstract `Axiom` base class. Each concrete axiom (e.g., `AtMostOneResourcePerAgent`, `RankEnvyFreeness`) inherits from this base and implements a `build()` method that generates the corresponding CNF clauses. This design enables:

- **Encapsulation**: Each axiom manages its own clause generation without direct global CNF manipulation
- **Extensible**: New fairness axioms are added by implementing the abstract `build()` method
- **Traceable**: Axioms track their clause indices, enabling later identification of which constraints contributed to an unsatisfiable state. This allows to easily trace back the clauses of a MUS to the responsible axiom.

The axioms implement the CNF translations detailed in the theoretical documentation.

SAT Solving and Satisfiability Testing

The `AllocationSatSolver` class wraps the SAT solver infrastructure, providing methods to test whether a set of axioms is satisfiable given the preference profile. It internally uses the `Solver` from the `pysat` library, accepting the CNF formula constructed by the axioms and returning satisfying assignments when solutions exist.

The solver operates on the CNF built by the `AxiomManager`, which registers all instantiated axioms and maintains the complete formula. This separation allows axioms to be instantiated incrementally while the manager ensures they are all incorporated into the final CNF.

Minimal Unsatisfiable Subset Computation

When no allocation satisfying all axioms exists, the `MUSComputer` class computes a Minimal Unsatisfiable Subset (MUS) using the MUSX algorithm from `pysat`. This identifies the minimal set of conflicting axioms, enabling diagnosis of why the allocation problem is infeasible.

The MUS is computed at the clause level but is then mapped back to the originating axioms through the clause-to-axiom mapping maintained by the `AxiomManager`. This back-mapping is critical for generating user-facing explanations.

Explanation Generation

The `MUSExplainer` and `AxiomExplainer` classes translate the mathematical constraints into natural language. Given a set of axioms from the MUS, the explainer generates structured descriptions of which constraints are in conflict. For example, an explanation might state: *"Agent 0 must be assigned at least one resource AND Agent 0 can be assigned to at most one resource"*, revealing the contradiction.

This addresses the goal of providing *local explanations*: agents can understand why a particular matching violates desirable fairness properties when no feasible solution exists.

Empirical Workflows

Two workflows were developed (`demo` directory):

- **Pipeline MUS**: Demonstrates the construction of axioms, SAT solving, MUS computation, and explanation generation for simple allocation problems.
- **Pipeline Disclosure**: Explores preference disclosure patterns for rank-envy-freeness via the $\mathrm{r_k}$-EF formulation.

8.2 Methodology and organization of the teamwork

Project overview

Figure 13 is an overview of the stages of the work for this project. We met with A.WILCZYNSKI and W.OUERDANE every two weeks on average to present them our work and get feedback.

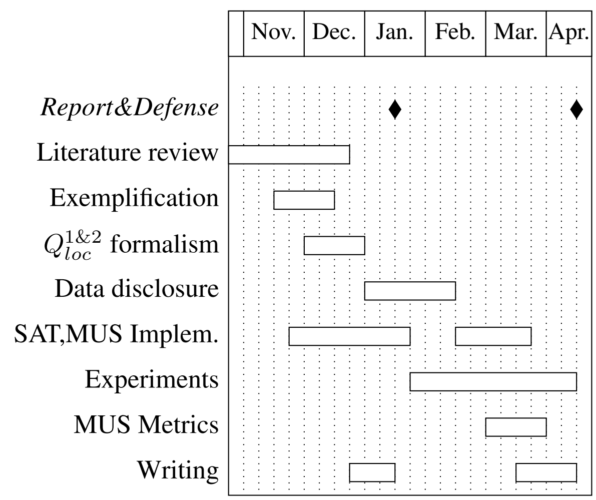

Figure 13: Project planning

Task splitting

Here is a summary of how the long term tasks were splitted :

Everyone started by reviewing literature, which we split between ourselves and we mutually presented ourselves the papers we had read. Then, M.GARNIER started working on exemplifying in different directions to determine the recurrent pattern in MUS explanations. S.LAMIK looking at what would explanations would look like. At the same time, A.FAURE started working on formalizing the prior questions, and arguments of an explanation.

This led to S.LAMIK starting to implement and validate the framework to work with SAT and MUSes to generate explanations experimentally. A.FAURE and M.GARNIER went onto working on the data disclosure caused by different parts of an explanation for Q_{loc}^1 and Q_{loc}^2.

M.GARNIER and S.LAMIK then went extending the framework to express the knowledge of possible preference profiles induced by an explanation, and A.FAURE worked on formalizing metrics to compare between MUS.

The last sprint of the project was dedicated to writing the final report and running experiments to infer some theoretical results.

8.3 Algorithms: Epistemic Leakage Measurement via MUS-Based Explanations under $\mathbf{r_k}$-EF

Algorithm 4: RUNSWEEP($\mathcal{N}$, $\mathcal{K}$, Φ, Σ)- Parallel experimental sweep.

Input : $\mathcal{N}$: set of agent counts;
$\mathcal{K}$: set of rank thresholds;
Φ: set of Mallows parameters;
Σ: set of seeds
Output: $\mathcal{R}$: result dictionary indexed by (N, k, φ)

`// Discard trivial pairs` ($RkEF(N) \equiv RkEF(N-1)$ `via serial dictatorship)`

$\mathcal{V} \leftarrow \left\{ (N,k) \mid N \in \mathcal{N},\ k \in \mathcal{K},\ k < N-1 \right\}$
foreach $(N,k) \in \mathcal{V}$ **do**
  foreach $\varphi \in \Phi$ **do**
    $B_{N,k,\varphi} \leftarrow$ EMPTYBUCKET(N,k,φ)
    `// Inner loop parallelised over seeds (process pool)`
    for $seed \in \Sigma$ **in parallel do**
      $r \leftarrow$ RUNSINGLESEED$(N,k,\varphi,seed)$ (Algorithm 3)
      $B_{N,k,\varphi} \leftarrow$ ACCUMULATE$(B_{N,k,\varphi},\ r)$
    $\mathcal{R}[(N,k,\varphi)] \leftarrow B_{N,k,\varphi}$
  end
end
return $\mathcal{R}$

Table 6: Summary of notation.

Symbol	**Definition**	**Symbol**	**Definition**
N	Number of agents (= objects)	$\mathcal{F}$	SAT formula of the Phase-1 solver
k	Rank threshold for $\text{RKEF}(k)$	$\mathcal{F}_{\text{bt}}$	SAT formula of the backtrack solver
$\varphi \in [0,1]$	Mallows dispersion parameter	M	Minimal Unsatisfiable Subset (MUS) of $\mathcal{F}'$
$seed$	Random seed (instance identifier)	Π_{prior}	Profile–allocation pairs compatible with i^*'s knowledge *before* MUS disclosure
$\mathcal{P}^{N,\varphi,\sigma}$	Ordinal profile sampled from Mallows(N, φ, σ)	Π_{post}	Idem *after* MUS disclosure
π_i	Total preference order of agent i over $[N]$	$\Pi_j^{\text{prior/post}}$	Marginal ordering set of agent j in $\Pi_{\text{prior/post}}$
$\sigma : [N] \to [N]$	Allocation (bijection)	$\Delta_{\text{abs}}^{(\sigma)}$	$\lvert\Pi_{\text{prior}}\rvert - \lvert\Pi_{\text{post}}\rvert$
i^*	Most dissatisfied agent	$\Delta_{\text{rel}}^{(\sigma)}$	$1 - \lvert\Pi_{\text{post}}\rvert \,/\, \lvert\Pi_{\text{prior}}\rvert$
o^*	Top-ranked object of i^*, i.e. $\pi_{i^*}(1)$	$\Delta_{\text{rel},j}^{(\sigma)}$	$1 - \lvert\Pi_j^{\text{post}}\rvert \,/\, \lvert\Pi_j^{\text{prior}}\rvert$

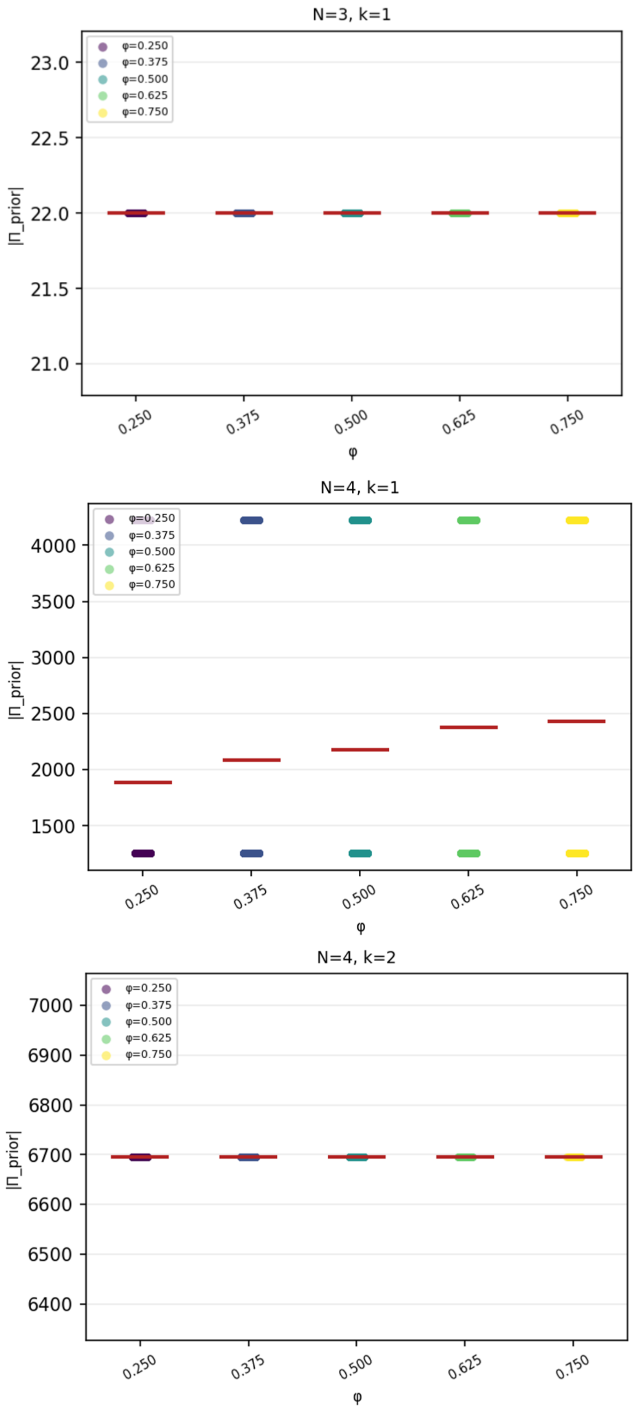

Figure 8: Strip plot of $|\Pi_{\text{prior}}|$ by φ for each (N, k) configuration. For $(N = 4,\ k = 1)$ two clusters are clearly visible; for $(N = 3,\ k = 1)$ the value is exactly 22; for $(N = 4,\ k = 2)$ it concentrates near 6700.

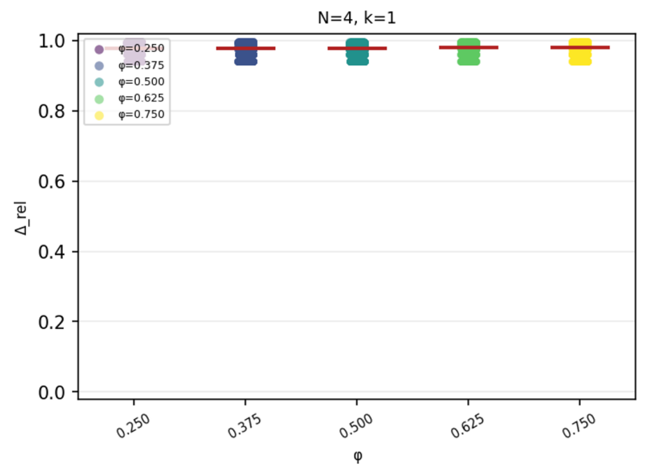

Figure 9: Strip plot of Δ_{rel} by φ for $(N = 4,\ k = 1)$, the within-cluster spread is visible.

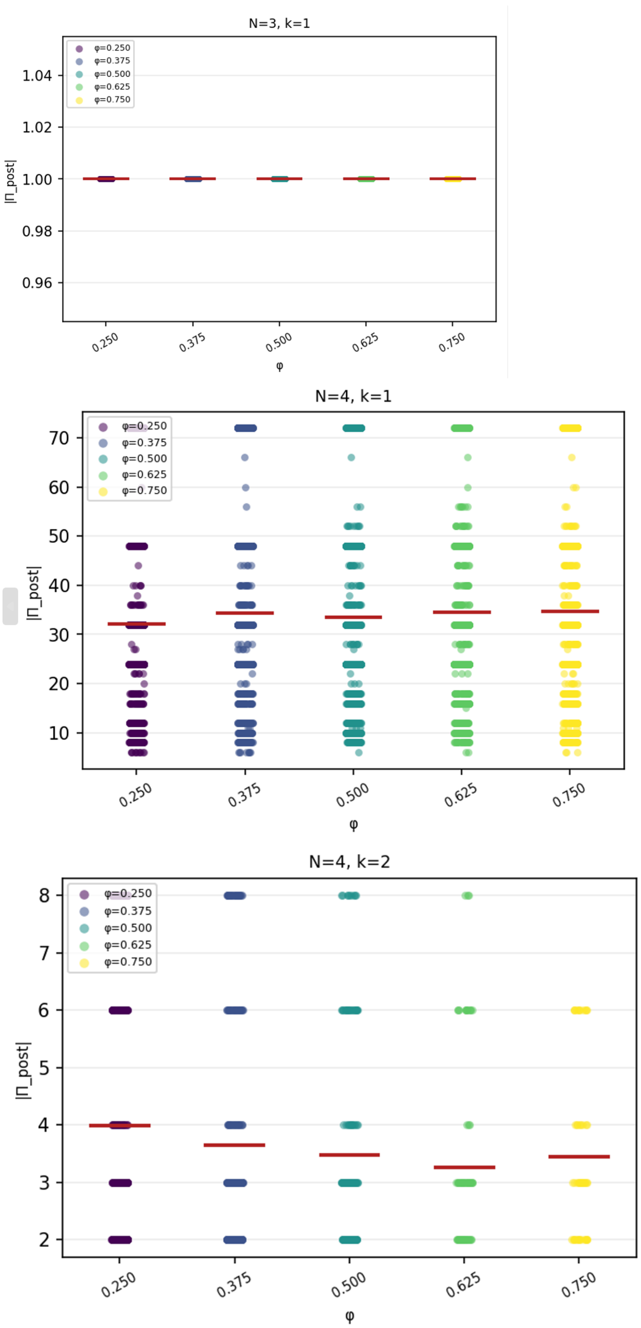

Figure 10: Strip plot of $|\Pi_{\mathrm{post}}|$ by φ for each (N, k) configuration. For $(N = 4,\ k = 2)$, the posterior takes only a few discrete integer values in $\{2, 3, 4, 6, 8\}$. For $(N = 3,\ k = 1)$, $|\Pi_{\mathrm{post}}| = 1$ exactly.

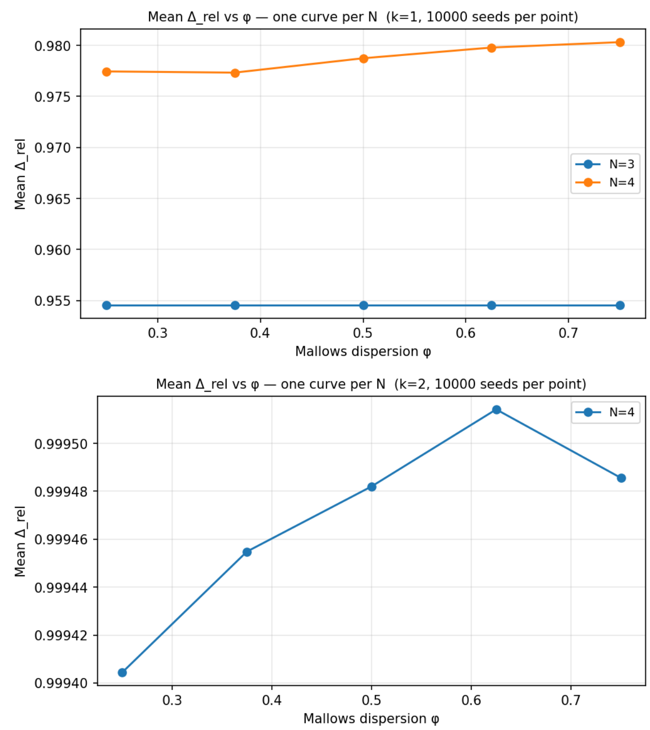

Figure 11: $\bar{\Delta}_{\mathrm{rel}}$ vs. φ for $N = 4$, $k \in \{1, 2\}$ (10 000 seeds per point). Both curves are nearly flat, confirming robustness to preference dispersion. The apparent peak at $\varphi = 0.625$ for $k = 2$ is not statistically robust ($n_{\mathrm{mus}} = 83$).

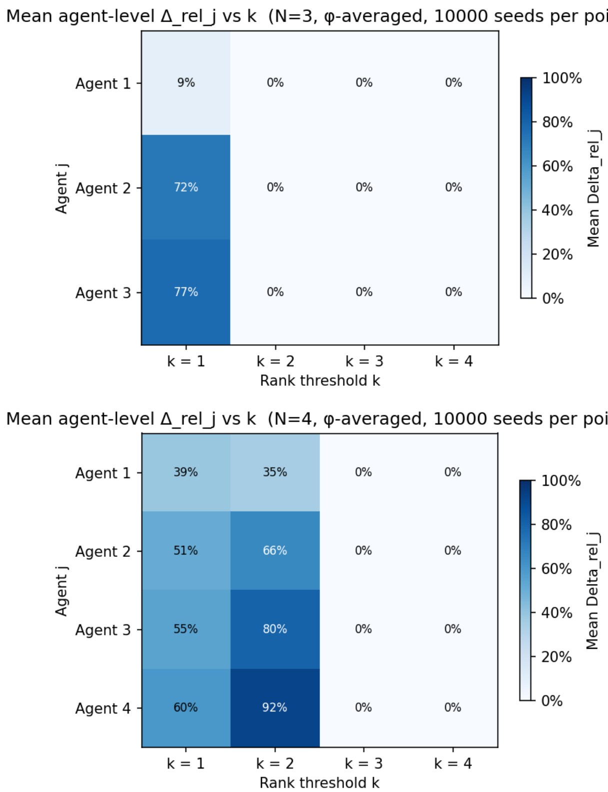

Figure 12: Mean per-agent $\bar{\Delta}_{\mathrm{rel},j}$ vs. rank threshold k (φ-averaged, 10 000 seeds per point). Left: $N = 3$. Right: $N = 4$. Disclosure is zero for $k \geq N - 1$ in all cases, and increases with agent rank (i.e., worse allocation) for $k < N - 1$.

Algorithm 5: AGGREGATEMETRICS($\mathcal{R}$, $\mathcal{N}$, $\mathcal{K}$, Φ)-Disclosure metric aggregation.

Input : $\mathcal{R}$: result dictionary (output of Algorithm 4); $\mathcal{N}, \mathcal{K}, \Phi$: parameter sets

Output: Aggregated metrics $\bar{\Delta}_{\mathrm{rel}}(N, k, \varphi)$, $\bar{\Delta}_{\mathrm{rel}}(N, k)$, $\tau_{\text{no-mus}}(N, k, \varphi)$, $\bar{\Delta}_{\mathrm{rel},j}(N, k)$

`// Per-bucket aggregation`

foreach $(N, k, \varphi) \in \mathcal{N} \times \mathcal{K} \times \Phi$ **do**

  Let $B = \mathcal{R}[(N, k, \varphi)]$, $S_{\mathrm{mus}} = \{seed \in \Sigma \mid r_{seed}$ is a MUS instance$\}$

  $\bar{\Delta}_{\mathrm{rel}}(N, k, \varphi) \leftarrow \frac{1}{|S_{\mathrm{mus}}|} \sum_{seed \in S_{\mathrm{mus}}} \Delta_{\mathrm{rel}}^{(seed)}$

  $\bar{\Delta}_{\mathrm{abs}}(N, k, \varphi) \leftarrow \frac{1}{|S_{\mathrm{mus}}|} \sum_{seed \in S_{\mathrm{mus}}} \Delta_{\mathrm{abs}}^{(seed)}$

  $\tau_{\text{no-mus}}(N, k, \varphi) \leftarrow \frac{n_{\text{no-mus}}}{n_{\mathrm{mus}} + n_{\text{no-mus}}}$

  for $j \in [N]$ **do**

    $\bar{\Delta}_{\mathrm{rel},j}(N, k, \varphi) \leftarrow \frac{1}{|S_{\mathrm{mus}}|} \sum_{seed \in S_{\mathrm{mus}}} \Delta_{\mathrm{rel},j}^{(seed)}$

  end

end

`// Aggregation over` Φ `(for fixed` (N, k)`)`

foreach $(N, k) \in \mathcal{V}$ **do**

  $\bar{\Delta}_{\mathrm{rel}}(N, k) \leftarrow \frac{1}{|\Phi|} \sum_{\varphi \in \Phi} \bar{\Delta}_{\mathrm{rel}}(N, k, \varphi)$

  $\bar{\tau}_{\text{no-mus}}(N, k) \leftarrow \frac{1}{|\Phi|} \sum_{\varphi \in \Phi} \tau_{\text{no-mus}}(N, k, \varphi)$

  for $j \in [N]$ **do**

    $\bar{\Delta}_{\mathrm{rel},j}(N, k) \leftarrow \frac{1}{|\Phi|} \sum_{\varphi \in \Phi} \bar{\Delta}_{\mathrm{rel},j}(N, k, \varphi)$

  end

end